

Physics Informed Priors for MLIPs with UQ

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1 Meta / abstract points

1. To propagate uncertainties up scales we need to capture MLIP errors at the level of MLIP parameters
2. However, the resultant model ensemble can be highly unphysical.
3. Here, we implement physics informed priors for linear MLIPs through inequality constraints
4. We provide an open source implementation within the POPS UQ scheme
5. For simple models, the constraints notably improve phonon stability
6. The final scheme provides robust max/min bounds while retaining physically meaningful model parameters.

2 Main points

1. To communicate that addressing holes in the potential energy surface is traditionally addressed by evaluating more DFT (via hyper-active learning, fine-tuning, etc.) and can be done less expensively
2. DFT stresses are not well converged so elastic properties are often incorrectly predicted by DFT
3. Non-repulsive core interactions for potentials are often observed, using our priors allows us to access high pressure conditions
4. Priors can be formulated based on experimental data
5. Inclusion of UQ via POPS and analysis of how this affects coverage of training data/test data

3 Method

1) linear ML potentials, extension to forces and Hessians / phonons 2) Stability criteria as linear inequalities 3) POPS method and need for constrained fits then the inequality extension of this paper

3.1 Linear MLIPs

An atom of specie s has a local atomic environment $\mathcal{R} = \{s, \mathbf{r}_1, s_1, \mathbf{r}_2, s_2, \dots\}$. Throughout this section, a and b indices denote atomic index and α and β denote Cartesian directions. We only apply Einstein summation to Cartesian indices. For an atomic configuration R the ACE energy is linear in the parameter vector $c \in \mathbb{R}^d$,

$$E(R; c) = \sum_{k=1}^d c_k B_k(R) = B(R) \cdot c, \quad (1)$$

with B_k the ACE basis functions. Because differentiation is linear, each basis function remains linear in its respective entry of c . This means that the force on an atom a is also linear in parameters and so is given by

$$\mathbf{F}_a = \sum_b \sum_{k=1}^d \frac{\partial B_k}{\partial \mathbf{r}_b} c_k = \sum_b \frac{\partial B}{\partial \mathbf{r}_b} c, \quad (2)$$

where b denotes the indices of the other atoms in the cell.

For cubic crystals, the volume is entirely parametrised by the lattice constant, a , which defines the unit cell size of the crystal whilst preserving the structure and symmetry. Thus, any homogenous and isotropic volumetric strain is described exactly by a change in the lattice constant for cubic crystal materials. The lattice constant fully describes the geometric distances of a given cubic crystal structure.

Thus we can describe the cell energy under such volumetric deformations as

$$E(a; c) = \sum_{k=1}^d B_k(a) c_k, \quad (3)$$

and the lattice constant which describes the minimum energy configuration crystal is then characterised exactly by

$$\frac{dE(a_0)}{da} = 0 \text{ eV } \text{\AA}^{-1}, \quad \frac{d^2 E(a_0)}{da^2} > 0 \text{ eV } \text{\AA}^{-2}. \quad (4)$$

Ensuring these constraints are obeyed while we enforce constraints that depend on geometry are essential as it removes the dependence on the lattice constant from the basis functions, linearising our further constraints.

Then the force-constant matrix, which is the second derivatives of the cell energy with respect to the positions of each atom in the cell

$$\Phi_{a\alpha, b\beta}(c) = \frac{\partial^2 E}{\partial r_{a\alpha} \partial r_{b\beta}} = \sum_{k=1}^d c_k \Phi_{a\alpha, b\beta}^{(k)}, \quad \Phi_{a,b}^{(k)} = \frac{\partial^2 B_k}{\partial \mathbf{r}_a \partial \mathbf{r}_b}, \quad (5)$$

is a third-order tensor that can be contracted with c , and is linear in c as a result. However, since the Hessian is the same for only a given geometry, if we have several models with different lattice constants our Hessian will still be implicitly dependent on c . To combat this, we can explicitly require each model we investigate to have a minimum at the same lattice constant a_0 , thus removing the dependence of the parameters from the basis functions of the Hessian.

From Bloch's theorem we know that $3N$ eigenvectors of the force constant matrix can be indexed by a wave-vector $\mathbf{k}_l$, $l \in [1, 3N]$ with Bloch structure

$$v_{l,a\alpha} = \exp(i\mathbf{k}_l \cdot \mathbf{x}_a) v_{l,\alpha}, \quad (6)$$

where $\mathbf{v}_l = [v_{l,1}, v_{l,2}, v_{l,3}]$. The original $3N \times 3N$ eigenvalue problem then reduces to N 3 eigenvalue problems

$$\tilde{\Phi}_{\alpha\beta}(\mathbf{k}_l) v_{l,\beta} = \lambda_l v_{l,\alpha}, \quad \tilde{\Phi}_{\alpha\beta}(\mathbf{k}_l) = \sum_b \frac{1}{\sqrt{mm_b}} \exp(i\mathbf{k}_l \cdot [\mathbf{x} - \mathbf{x}_b]) \Phi_{\alpha,b\beta}(c). \quad (7)$$

For each eigenvector, we can take the dot product between the transpose of the eigenvector on both sides of the equation

$$\begin{aligned} \tilde{\Phi}_{\alpha\beta}(\mathbf{k}_l) v_{l,\beta} &= \lambda_l v_{l,\alpha} \\ v_{l,\alpha}^* \tilde{\Phi}_{\alpha\beta}(\mathbf{k}_l) v_{l,\beta} &= \lambda_l v_{l,\alpha}^* v_{l,\alpha} \\ v_{l,\alpha}^* v_{l,\beta} \sum_{k=1}^d \tilde{\Phi}_{\alpha\beta}^k(\mathbf{k}_l) c^k &= \lambda_l v_{l,\alpha}^* v_{l,\alpha} \end{aligned} \quad (8)$$

Because our dynamical matrix is Hermitian the eigen-vectors are orthogonal, and so after normalisation the Euclidean norm of each eigen-vector is 1, and so

$$v_{l,\alpha}^* v_{l,\beta} \sum_{k=1}^d \tilde{\Phi}_{\alpha\beta}^k(\mathbf{k}_l) c_k > 0 \Leftrightarrow \lambda_l > 0. \quad (9)$$

Eq. 9 thus is an exactly linear form of the real phonon energy constraint.

However, during constrained optimization the parameter vector $\mathbf{c}$ is subject to change. Due to the eigen-vectors' non-linear dependence on $\mathbf{c}$ via the dynamical matrix we can not enforce that they do not change. So, after the constrained optimization finishes we must check that the new dynamical matrix has positive eigenvalues. If it doesn't then we can simply reconstruct the constraints and re-constrain with these new added constraints.

3.2 POPS-Regression

These physical expressions serve as our priors to the pointwise optimal parameter set regression (POPS-Regression) uncertainty quantification method. Unlike Bayesian methods, POPS-Regression is model-misspecification aware and guarantees that the posterior distribution does not shrink to a Dirac-delta function as $N_{training} \rightarrow \infty$.

This guarantee is provided by calculating $N_{training}$ corrections to the RLS solution, where each correction interpolates exactly one data point in the training data set, while minimizing the loss function with the rest of the data. The analytical form of this for a given data point, i , is

$$\delta c_i = \frac{y_i - B_i \cdot c_{RLS}}{B_i^T P^{-1} B_i} P^{-1} B_i. \quad (10)$$

From the corrections we take the singular value decomposition to get the eigen-vectors, which we apply the inverse of the the corrections. With the transformed point-cloud of corrections, we fit a hyper-cube and draw uniform samples.

We now have several scenarios to which we provide solutions:

1. At least one POPS correction fit leads to unphysical predictions of quantities of interests and so do samples from the hyper-cube in this region of parameter space
2. The samples from the hypercube have unphysical predictions because the hypercube extends the valid region for sampling, even if the point-cloud of corrections are all physical
3. The least squares solution itself is unphysical

3.3 Enforcing Constraints

$$\text{Minimize} \quad \|Bc - y\|_2^2 + \|\Gamma c\|_2^2 \quad (11)$$

Enforcing linear constraints during training can be achieved by utilising the quadratic programming with linear constraints framework. This can be described mathematically by

$$\begin{aligned} &\text{minimize} \quad \frac{1}{2} c^T P c + q^T c \\ &\text{subject to} \quad l \leq A c \leq u, \end{aligned} \quad (12)$$

where $c \in \mathbb{R}^d$ is the model parameters that we are minimizing. The matrix $P \in \mathbb{S}_+^d$ is the regularised Gram matrix given by $P := B^T B + \Gamma^T \Gamma$, and $q \in \mathbb{R}^d$ is given by $q := B^T y$.

To tackle POPS correction fits predicting unphysical predictions we can loop over the unphysical POPS. Since the eigenvectors of our matrix can change as we change our parametrisation, we require the require the constraint consistency loop for each constrained POPS calculation. We can guarantee that the constrained optimization problem does not become infeasible as a result of the Rayleigh-Ritz Theorem [Horn'Johnson'1985] which for a Hermitian matrix A states

$$\lambda_{min} \leq \frac{x^\dagger A x}{x^\dagger x} \leq \lambda_{max}, \quad \|x\|_2^2 \neq 0. \quad (13)$$

Algorithm 1 Constrained POPS

Require: $B, \Gamma, y, c_{RLS}, a_{RLS}$ **Ensure:** $\tilde{\Phi}(\mathbf{q}) \cdot c_{POPS} \succeq 0 \forall \mathbf{q} \in q_{bandpath}$

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1: Construct  $\tilde{\Phi}(\mathbf{q}), B'(a_{RLS}), B''(a_{RLS})$ 
2: for  $\mathbf{q} \in q_{bandpath}$  do
3:    $U, \Lambda, V = \text{SVD}(\tilde{\Phi}(\mathbf{q}) \cdot c_{POPS, i})$ 
4:    $\tilde{\Phi}_{constraint} \leftarrow \text{vcat}(\tilde{\Phi}_{constraint}, V^T \tilde{\Phi}(\mathbf{q}) V)$ 
5: end for
6: for  $i = 1 : N$  do
7:   while  $\tilde{\Phi}(\mathbf{q}) \cdot c_{POPS, i} \not\succeq 0 \forall \mathbf{q} \in q_{bandpath}$  do
8:      $A \leftarrow \text{vcat}(B_{i,:}, B'(a_{RLS}), B''(a_{RLS}), \tilde{\Phi}_{constraint})$ 
9:      $l \leftarrow \text{vcat}(Y_i, 0, \delta_{tol}, \delta_{tol})$ 
10:     $u \leftarrow \text{vcat}(Y_i, 0, \infty, \infty)$ 
11:     $c_{POPS, i} = \min \|Bc - y\|_2^2 + \|\Gamma c\|_2^2$  subject to  $l \leq Ac \leq u$ 
12:    for  $\mathbf{q} \in q_{bandpath}$  do
13:       $U, \Lambda, V = \text{SVD}(\tilde{\Phi}(\mathbf{q}) \cdot c_{POPS, i})$ 
14:      if  $V^T \tilde{\Phi}(\mathbf{q}) V \cdot c_{POPS, i} < 0$  then
15:         $\tilde{\Phi}_{constraint} \leftarrow \text{vcat}(\tilde{\Phi}_{constraint}, V^T \tilde{\Phi}(\mathbf{q}) V)$ 
16:      end if
17:    end for
18:  end while
19:   $c_{POPS} \leftarrow \text{hcat}(c_{POPS}, c_{POPS, i})$ 
20: end for
21: return  $c_{POPS}$ 
```

Since we are constraining only a lower bound on the eigenvalues, combining any set of eigen-vectors which have non-zero Euclidean norm to our set of constraints will always be feasible. Thus, we can indefinitely add more constraints corresponding to the parameter space we have already explored and we construct a set of half-spaces that define the positive semi-definite cone of dynamical matrices that within our model's parameter space.

Standard POPS-Regression then builds a hypercube around the set of point-wise corrections and uniformly samples. To avoid sampling unphysical regions of parameter space accessed by the corners and surfaces of the hypercube, we implement rejection sampling. This protocol is to draw a sample and check whether the sampled parameters obey the constraints.

4 Results

4.1 Constraining Real Phonons

The first result that we present is the enforcement of real-valued phonon energies for Al at 0K for two ACEpotentials models POPS uncertainty ensembles. Fig. 1

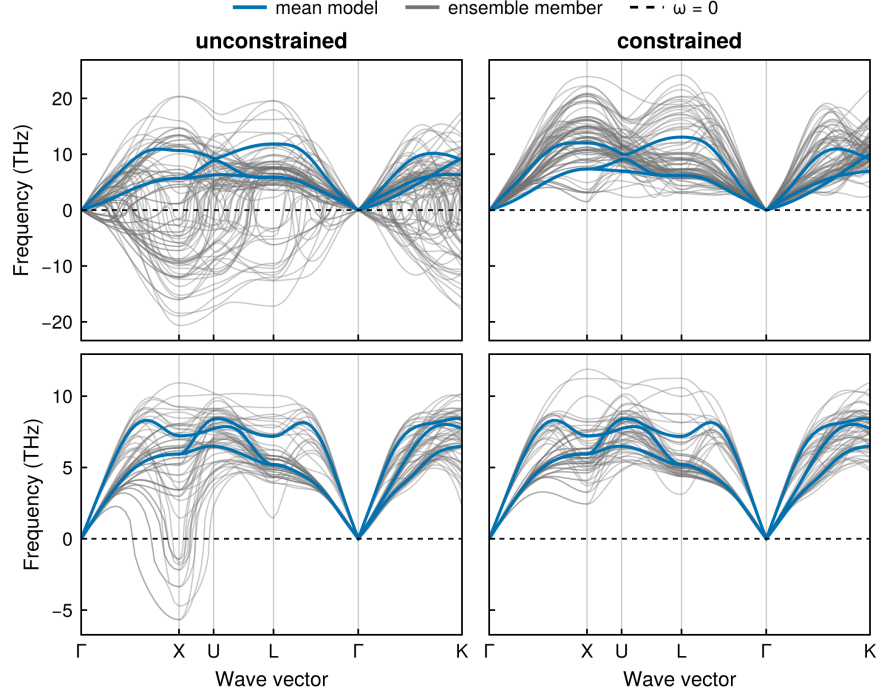


Figure 1: Comparison of Aluminium 0K phonon bands before and applying constraints for phonon positivity across the bandpath at the equilibrium volume.

displays the effects of using constrained POPS compared to the standard POPS algorithm.

The smaller ACE model used in the top panel of Fig. 1 was trained on 7893 configurations, giving a total of 146958 independent observations. The model goes up to the third body-order, has a cut-off radius of $6\text{\AA}$, a maximum polynomial degree of 12, giving 91 parameters in total. The larger model we use in the bottom panel of Fig. 1 shares the same training dataset and hyperparameters except for a larger maximum polynomial degree of 16, resulting in 684 parameters.

However, in practice we care about lattice stability at temperatures according to experimental observations. Since the only change to Eq. 5 is the geometry in the Hessian, the constraint is still linear in parameters and we can enforce positivity easily.

After picking a subset of data-points to perform constrained POPS at multiple lattice constants and rejection sampling the constrained hypercube we compare to standard POPS in Fig. 2. The two ensemble members shown were picked because they contain the smallest frequencies squared across the whole band-path out of each ensemble.

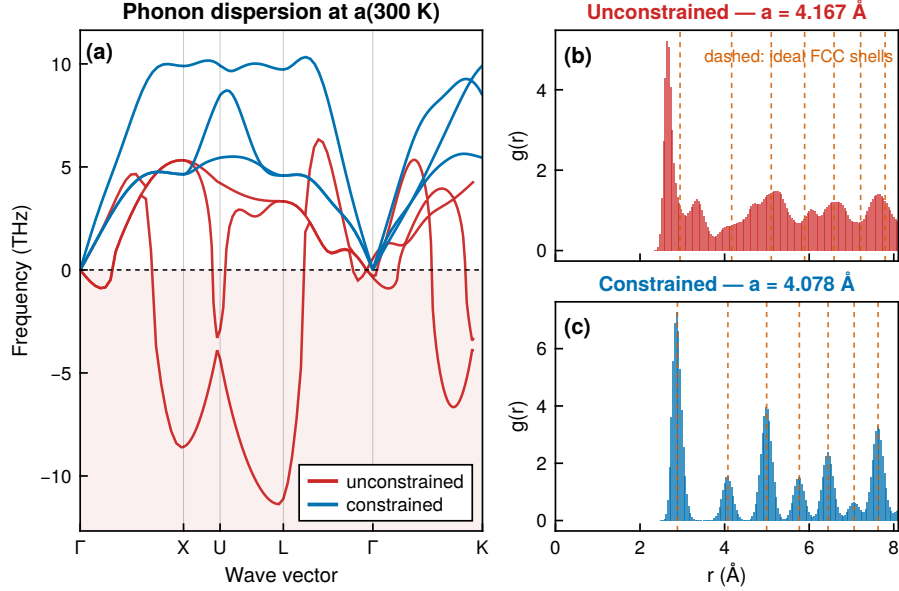


Figure 2: a) Phonon bands of two Al ACE models at lattice constant predicted by 300K NPT simulation using a naive POPS hypercube sample (shown in red) and a sample from rejection sampling the hypercube built from a subsample of the POPS delta forest that had been constrained phonon positivity at multiple volumes. b) and c) compare the radial distribution functions of the potentials from the two methods.

We also run Langevin dynamics at several temperatures and 0Pa to observe the change in cell volume and, by extension, lattice constant. Fig. 3 suggests that the constrained model leaves FCC at 900K, which is thermodynamically inaccurate for pure Al at that temperature. However, the thermal fluctuations of the MD run deviate too far from 900K for the potential to remain physical. This not only shows the flexibility of even a 91 parameter ACEpotential but also demonstrates the impact that linear inequality constraints can have on dynamical simulations.

4.2 Constraining Repulsive Core for Tungsten

We also conducted a study on a tungsten ACE model with 1829 parameters. We observed that under extreme compression the mean potential predicted the atoms to leave BCC sites, to collapse onto each other, and thus the simulation cell to collapse inwards too. We observed this phenomenon in an NPT simulation at 5TPa and 2200K. While these simulation conditions are extreme they still highlight the capabilities of enforcing physics in potentials using our method. With no extra data, we were able to convert our potential from useless in a

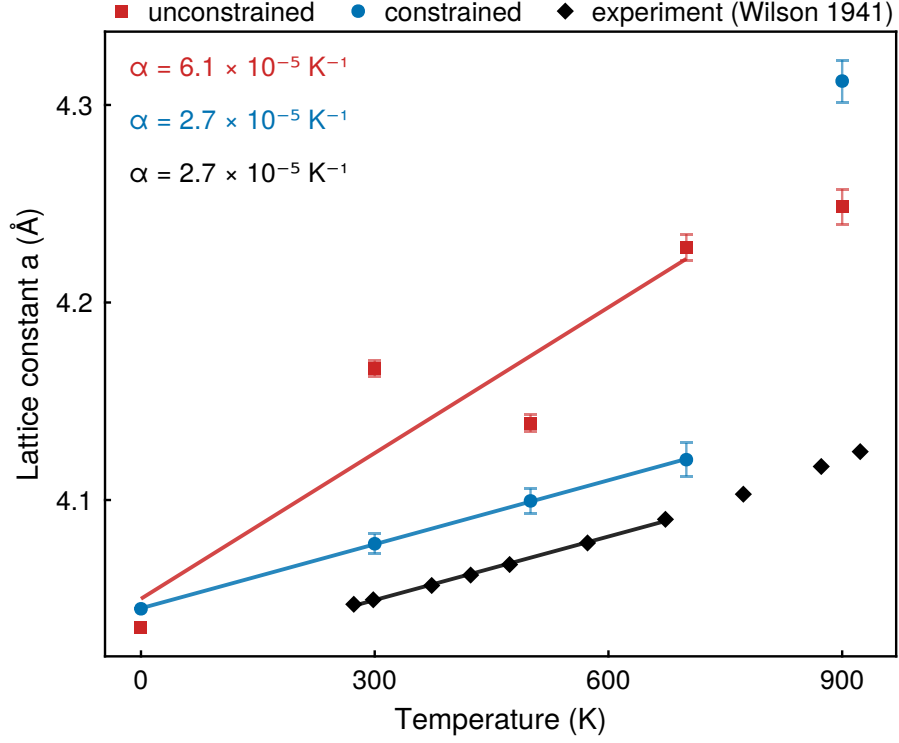


Figure 3: Comparison of the two predictions of lattice constants at different temperatures by the unconstrained and constrained AI ACE models. The unconstrained model leaves FCC phase at 300K and above whereas the constrained model predicts linear behaviour of the lattice constant, and that the lattice remains FCC until 900K. The black data also displays experimental measurements.

data sparse area of configuration space to insightful. Access to these extreme conditions of the pure bulk crystal also facilitates studies of defects in these conditions.

5 Discussion

References

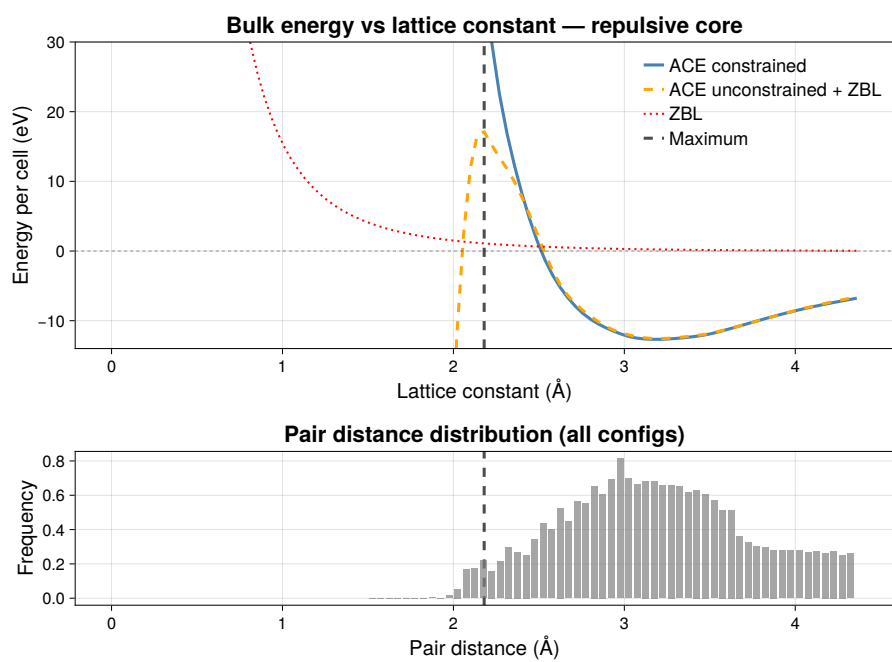


Figure 4: Bulk Tungsten cell energies for different lattice constants, with histogram detailing the distribution of pair distances in the training data set.